



005246 Signals and Systems

Credit Hours: 3+0 (3)

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Fourier Series

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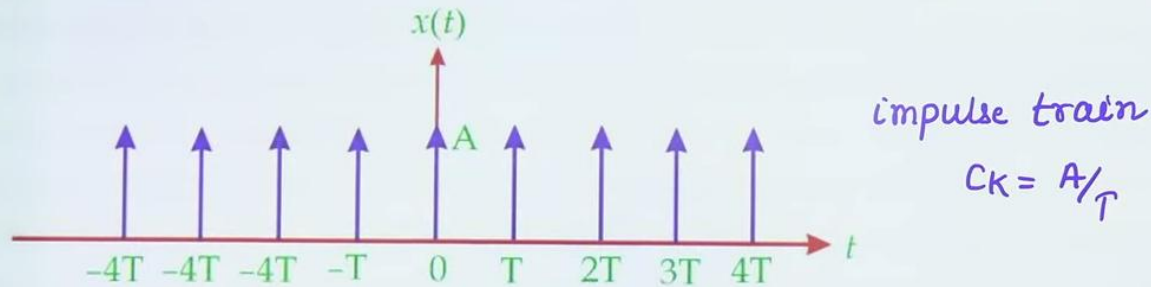
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Fourier Series

- ✓ Review of Basics of Vectors and Orthogonal Signals
- ✓ Definition of Continuous-time Fourier Series (CTFS) for Periodic Signals
- ✓ Trigonometric CTFS
 - ✓ Fourier Series Coefficients
 - ✓ Polar/Harmonic Form of Trigonometric CTFS
 - Effect of Symmetry on Fourier Series Coefficients
- ✓ Exponential CTFS
 - ✓ Magnitude and Phase Spectrum
- ✓ Relation between Trigonometric and Exponential CTFS Coefficients
- ✓ Properties of CTFS
 - Discrete-time Fourier Series (DTFS)

Examples

Determine Fourier series coefficients of following signals



If $x(t) \xrightarrow{FS} c_n = \left(\frac{1}{3}\right)^n u[n] \quad \forall n \in I$

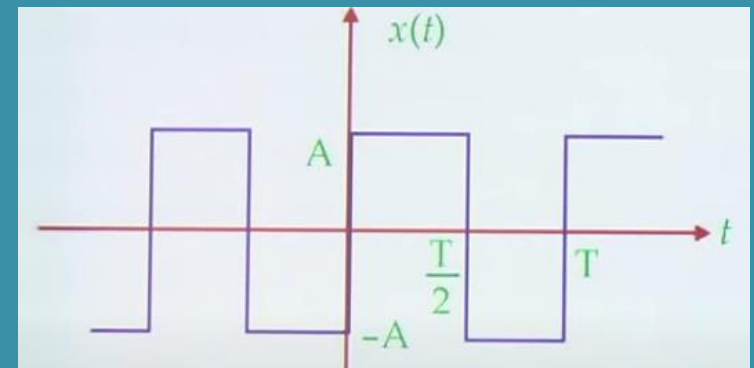
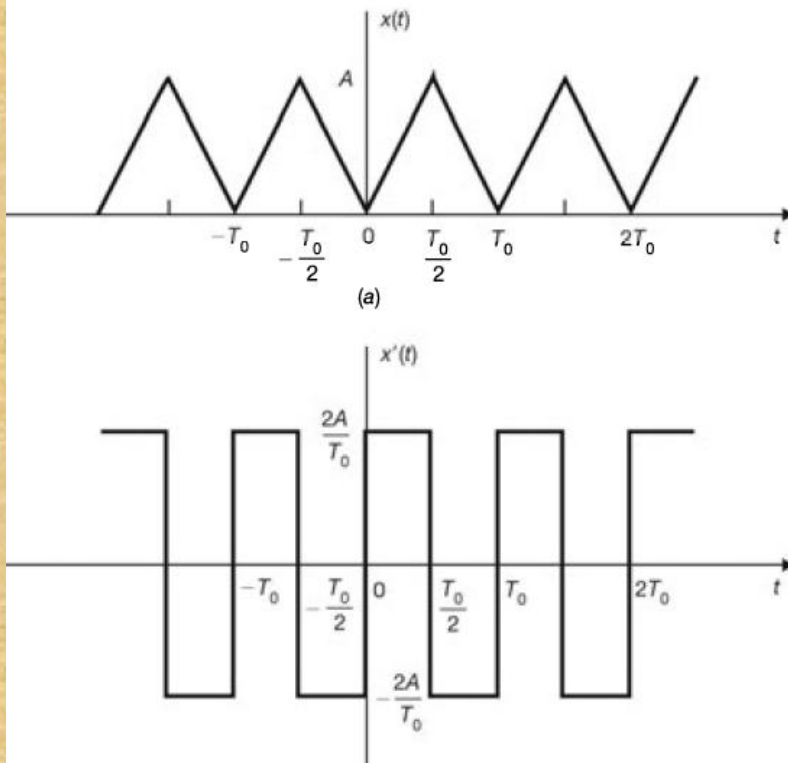
Determine (i) P_x (ii) $\{x(t)\}_{t=0}$

$$\begin{aligned} \text{(i)} \quad P_x &= \sum_{n=-\infty}^{\infty} |c_n|^2 = \sum_{n=0}^{\infty} \left[\left(\frac{1}{3}\right)^n \right]^2 \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{9}\right)^n = \frac{1}{1 - 1/9} = 9/8 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad x(t) \Big|_{t=0} &= \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} \Big|_{t=0} \\ &= \sum_{k=-\infty}^{\infty} c_k = \sum_{k=0}^{\infty} \left(\frac{1}{3}\right)^k = \frac{1}{1 - 1/3} = 3/2 \end{aligned}$$

Examples

Consider the triangular wave $x(t)$ Using the differentiation property, find the complex exponential Fourier series of $x(t)$



$$C_n = \frac{A}{jn\pi} [1 - e^{-jn\pi}]$$

$$e^{-jn\pi} = (-1)^n = \begin{cases} 1, & n = \text{even} \\ -1, & n = \text{odd} \end{cases}$$

$$C_n = \frac{2A}{jn\pi}, \quad n = \text{odd}, \quad C_n = 0, \quad n = \text{even}$$

$$c_0 = \frac{A}{2}$$

$$jn\omega_0 c_n = \frac{4A}{jn\pi T_0} \quad \forall n = \text{odd},$$

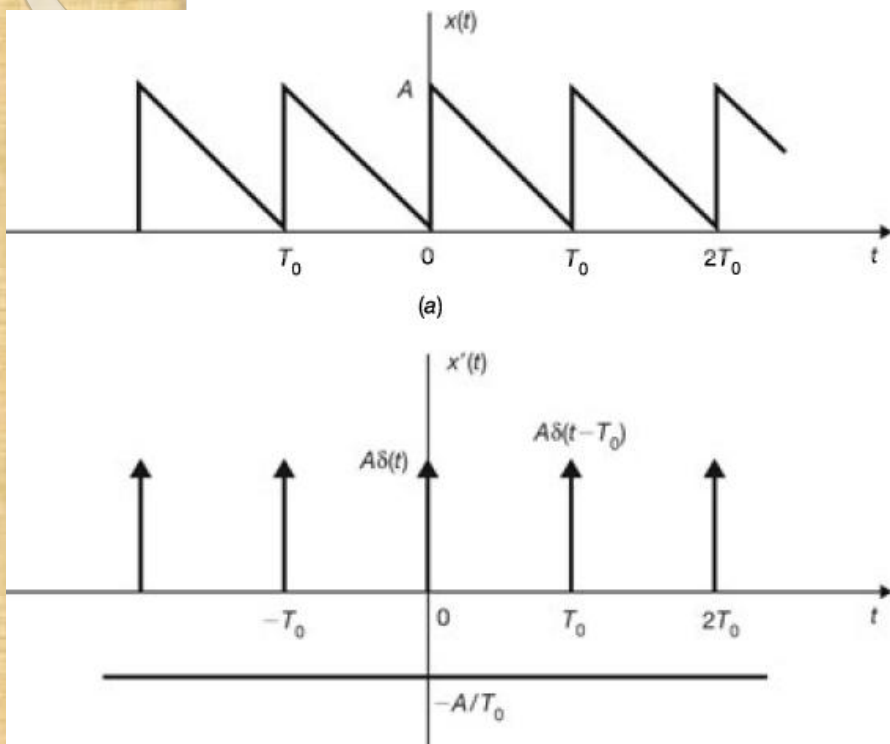
$$c_n = 0 \quad \forall n = \text{even}$$

$$c_n = -\frac{2A}{n^2\pi^2} \quad \forall n = \text{odd},$$

$$c_n = 0 \quad \forall n = \text{even}$$

Examples

Consider the triangular wave $x(t)$ Using the differentiation property, find the complex exponential Fourier series of $x(t)$



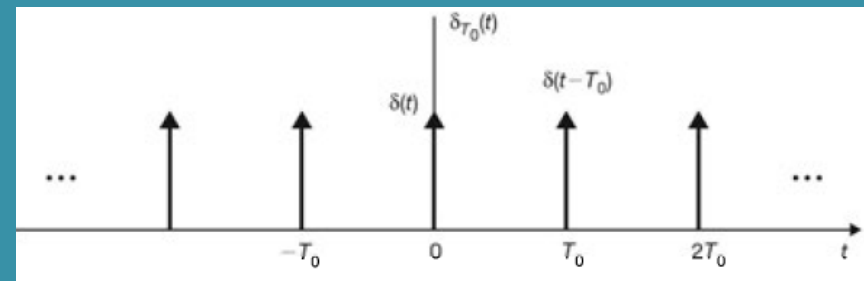
$$c_0 = -\frac{A}{2}$$

$$jn\omega_0 c_n = \frac{A}{T_0} \forall n$$

$$c_n = \frac{A}{jn2\pi} \forall n$$

Note: FS coefficient of Impulse Train

$$\delta_{T_0}(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_0)$$



$$c_n = \frac{1}{T_0}$$

Discrete-time FS

- It is a tool for analysis of infinite duration discrete-time periodic signals.
- DTFS is the only Fourier representation having finite number of discrete components in both time and frequency domains.
- General periodic signal $f[n]$ can be represented as a linear combination of harmonically related complex exponentials

$$f[n] = \sum_{k=0}^{N-1} c_k e^{jk\omega_0 n}$$

Synthesis Equation

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk\omega_0 n}$$

Analysis Coefficient

N is the period of DT signal

$\omega_0 = \frac{2\pi}{N}$ is fundamental frequency

Difference b/w Continuous-time and Discrete-time Fourier Series

- DTFS is finite series (has finite numbers of coefficients), while CTFS is an infinite series.
- Since in discrete domain there are finite numbers in a sequence. There is no convergence issues. No Dirichlet condition required.
- DTFS coefficients are periodic while CTFS coefficients are aperiodic.

Properties of DTFS

TABLE 3.2 PROPERTIES OF DISCRETE-TIME FOURIER SERIES

Property	Periodic Signal	Fourier Series Coefficients
	$\left. \begin{array}{l} x[n] \\ y[n] \end{array} \right\} \begin{array}{l} \text{Periodic with period } N \text{ and} \\ \text{fundamental frequency } \omega_0 = 2\pi/N \end{array}$	$\left. \begin{array}{l} a_k \\ b_k \end{array} \right\} \begin{array}{l} \text{Periodic with} \\ \text{period } N \end{array}$
Linearity	$Ax[n] + By[n]$	$Aa_k + Bb_k$
Time Shifting	$x[n - n_0]$	$a_k e^{-jk(2\pi/N)n_0}$
Frequency Shifting	$e^{jM(2\pi/N)n} x[n]$	a_{k-M}
Conjugation	$x^*[n]$	a_{-k}^*
Time Reversal	$x[-n]$	a_{-k}
Time Scaling	$x_{(m)}[n] = \begin{cases} x[n/m], & \text{if } n \text{ is a multiple of } m \\ 0, & \text{if } n \text{ is not a multiple of } m \end{cases}$ (periodic with period mN)	$\frac{1}{m} a_k$ (viewed as periodic with period mN)
Periodic Convolution	$\sum_{r=\langle N \rangle} x[r]y[n-r]$	$Na_k b_k$
Multiplication	$x[n]y[n]$	$\sum_{l=\langle N \rangle} a_l b_{k-l}$
First Difference	$x[n] - x[n-1]$	$(1 - e^{-jk(2\pi/N)})a_k$
Running Sum	$\sum_{k=-\infty}^n x[k]$ (finite valued and periodic only) if $a_0 = 0$	$\left(\frac{1}{(1 - e^{-jk(2\pi/N)})} \right) a_k$
Conjugate Symmetry for Real Signals	$x[n]$ real	$\begin{cases} a_k = a_{-k}^* \\ \Re\{a_k\} = \Re\{a_{-k}\} \\ \Im\{a_k\} = -\Im\{a_{-k}\} \\ a_k = a_{-k} \\ \angle a_k = -\angle a_{-k} \end{cases}$
Real and Even Signals	$x[n]$ real and even	a_k real and even
Real and Odd Signals	$x[n]$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals	$\begin{cases} x_e[n] = \mathcal{E}\{x[n]\} & [x[n] \text{ real}] \\ x_o[n] = \mathcal{O}\{x[n]\} & [x[n] \text{ real}] \end{cases}$	$\begin{cases} \Re\{a_k\} \\ j\Im\{a_k\} \end{cases}$
Parseval's Relation for Periodic Signals		
$\frac{1}{N} \sum_{n=\langle N \rangle} x[n] ^2 = \sum_{k=\langle N \rangle} a_k ^2$		

Examples

Complete Fourier series coefficient of $x[n] = \sin [0.2\pi n]$

$$\omega_0 = 0.2\pi$$

$$x[n] = \sin(0.2\pi n) = \frac{e^{j0.2\pi n} - e^{-j0.2\pi n}}{2j}$$

$$= \frac{1}{2j} e^{j\omega_0 n} - \frac{1}{2j} e^{-j\omega_0 n} = \sum c_k e^{-jk\omega_0 n}$$

$\downarrow$ $\downarrow$

$K=1$ $K=-1$

$$c_1 = \frac{1}{2j}$$

Examples

Complete power of given signal by Parseval's relationship.

$$x[n] = \cos\left(\frac{n\pi}{5}\right) + \sin\left(\frac{n\pi}{6}\right)$$

$$N_1: 2\pi/\pi/5 = 10$$

$$N_2 = 2\pi/\pi/6 = 12$$

$$N = \text{LCM}(10, 12) = 60, \quad \omega_0 = 2\pi/60 = \pi/30$$

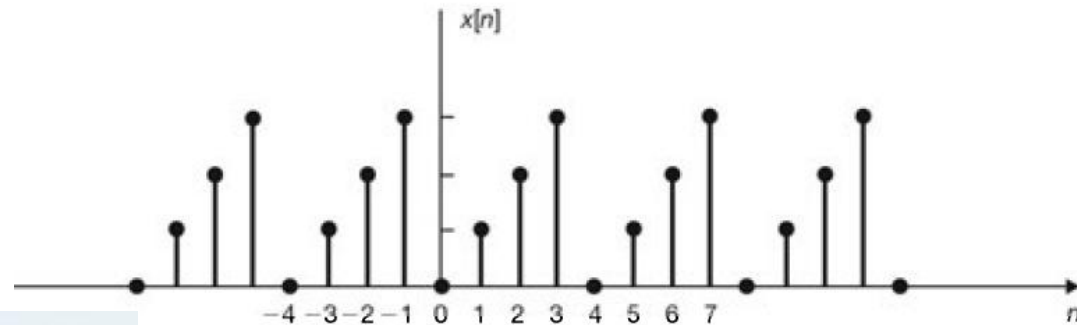
$$= \frac{1}{2} e^{jn\pi/5} + \frac{1}{2} e^{-jn\pi/5} + \frac{1}{2j} e^{jn\pi/6} - \frac{1}{2j} e^{-jn\pi/6}$$

$$= \frac{1}{2} e^{jn6\pi/30} + \frac{1}{2} e^{jn(-6)\pi/30} + \frac{1}{2j} e^{jn(5)\pi/30} - \frac{1}{2j} e^{jn(-5)\pi/30}$$

$$P = \sum |c_n|^2 = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2$$
$$= 1$$

Examples

Determine the Fourier coefficients for the periodic sequence $x[n]$ shown in Fig. 6-7.



$$N_0 = 4$$

$$\omega_0 = \frac{2\pi}{N_0} = \pi/2$$

$$C_k = \frac{1}{N} \sum_{\langle n \rangle} x[n] e^{-j k \omega_0 n}$$

$$= \frac{1}{N} \sum_{n=0}^3 x[n] e^{-j k \pi/2 n}$$

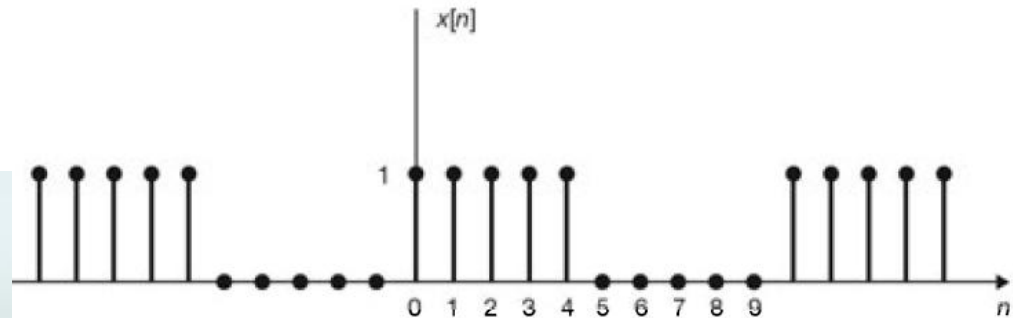
$$= \frac{1}{4} \left[x[0] e^{-j k \pi/2 (0)} + x[1] e^{-j k \pi/2 (1)} \right. \\ \left. + x[2] e^{-j k \pi/2 (2)} + x[3] e^{-j k \pi/2 (3)} \right]$$

$$\left\{ \dots, \frac{3}{2}, -\frac{1+j}{2}, -\frac{1}{2}, -\frac{1-j}{2}, \frac{3}{2}, \dots \right\}$$

↑

Examples

Determine the Fourier coefficients for the periodic sequence $x[n]$ shown in Fig.



$$N_0 = 10 \quad \omega_0 = 2\pi/N_0 = \pi/5$$

$$C_k = \frac{1}{10} \sum_{n=0}^9 x[n] e^{-jk\omega_0 n}$$

$$= \frac{1}{10} \sum_{n=0}^4 1 \cdot e^{-jk\omega_0 n}$$

$$= \frac{1}{10} \times \frac{1 - (e^{-jk\omega_0})^5}{(1 - e^{-jk\omega_0})}$$

$$C_k = \frac{1}{10} \left(\frac{1 - e^{-j5k\pi/5}}{1 - e^{-jk\pi/5}} \right)$$

$$C_k = \frac{1}{10} \left(\frac{1 - e^{-jk\pi}}{1 - e^{-jk\pi/5}} \right)$$

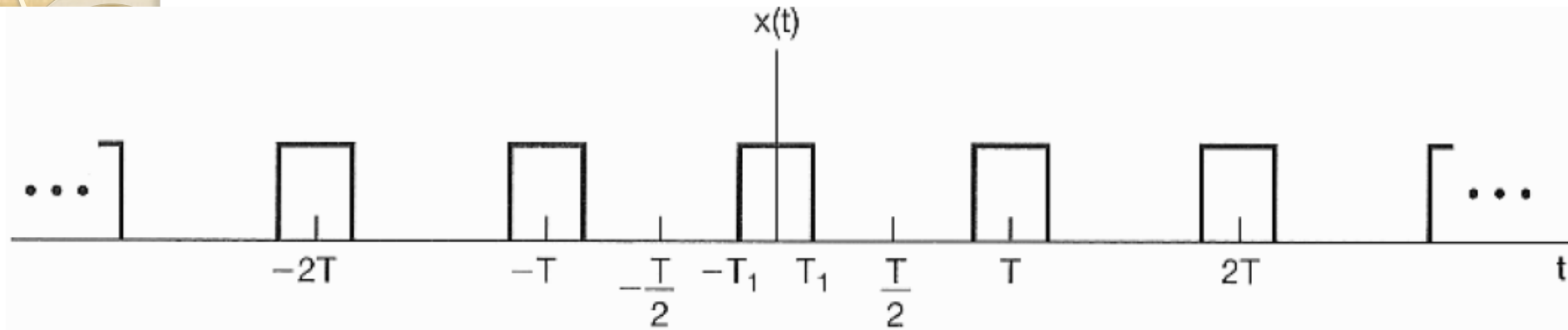
Fourier Transform

- Definition of Continuous-time Fourier transform (CTFT)
 - Relationship between $X(\omega)$ and $X(f)$
 - Existence of Fourier Transform
 - Magnitude and Phase Spectrum
- Common Fourier Transforms
- Properties of CTFT
- Characterization of LTI Systems
- Discrete-time Fourier Transform (DTFT)

Continuous-time Fourier Transform

- Fourier series provide an extremely useful representation for periodic signals.
- Unfortunately, this signal representation can only be used for periodic signals, since a Fourier series is inherently periodic.
- Often, however, we need to deal with signals that are not periodic.
- A more general tool than the Fourier series is needed in this case.
- The Fourier transform can be used to represent both periodic and aperiodic signals.
- Since the Fourier transform is essentially derived from Fourier series through a limiting process, the Fourier transform has many similarities with Fourier series.

Continuous-time Fourier Transform

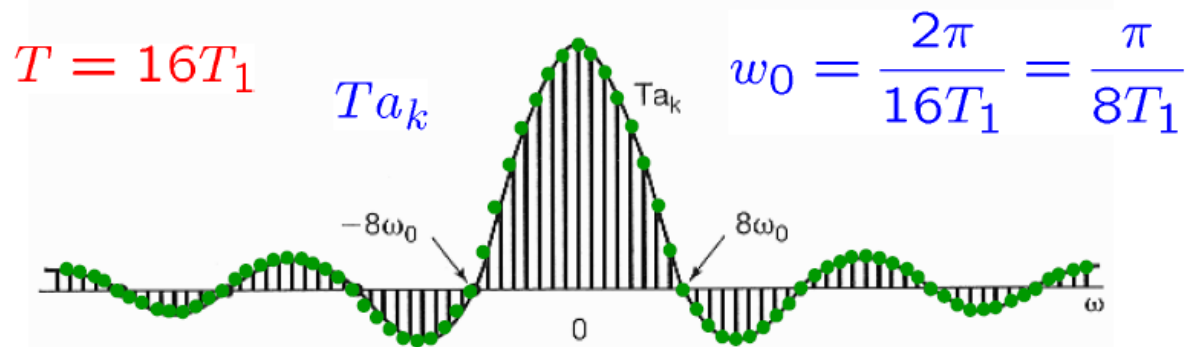
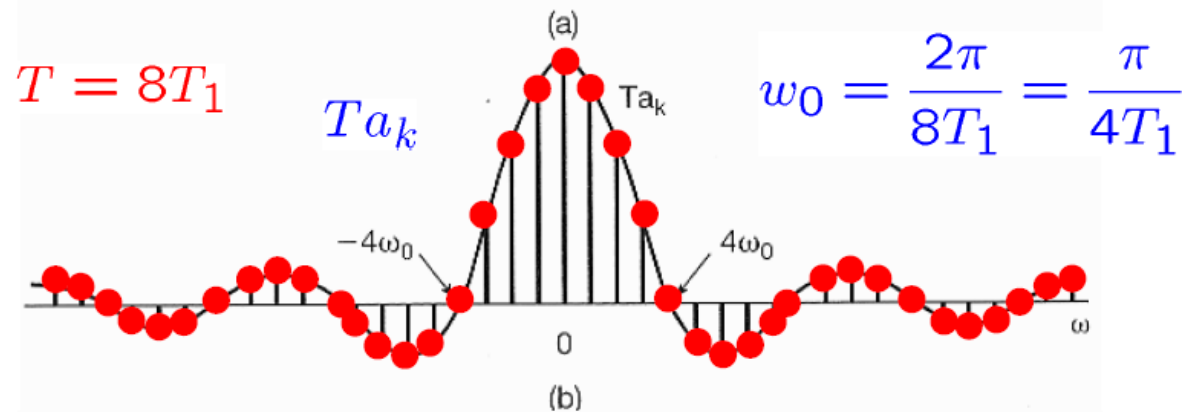
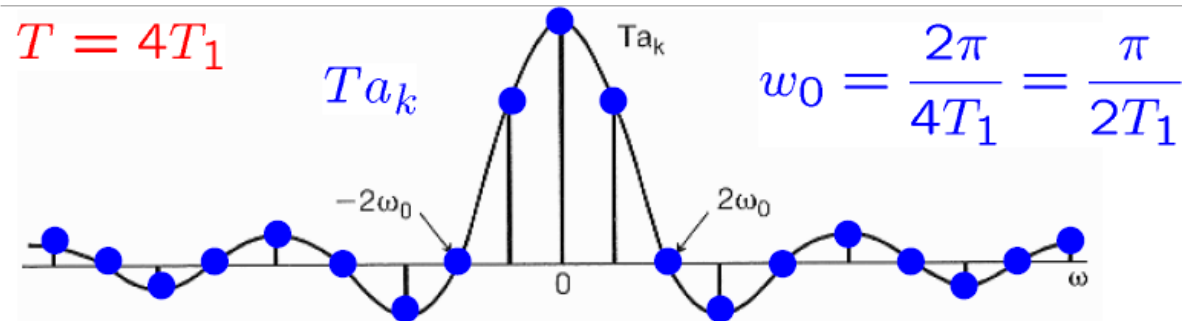


$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$

$$a_k = \frac{2 \sin(k\omega_0 T_1)}{k\omega_0 T}$$

Fourier series coefficients

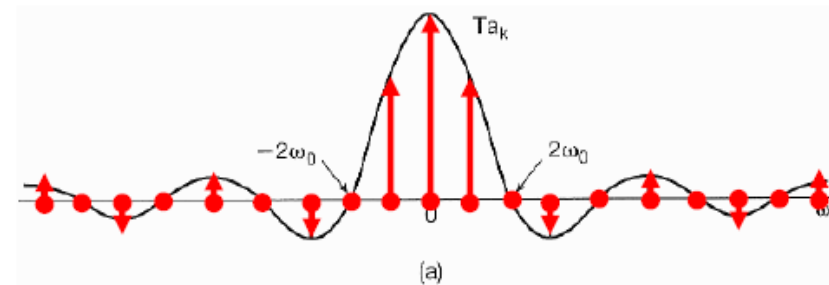
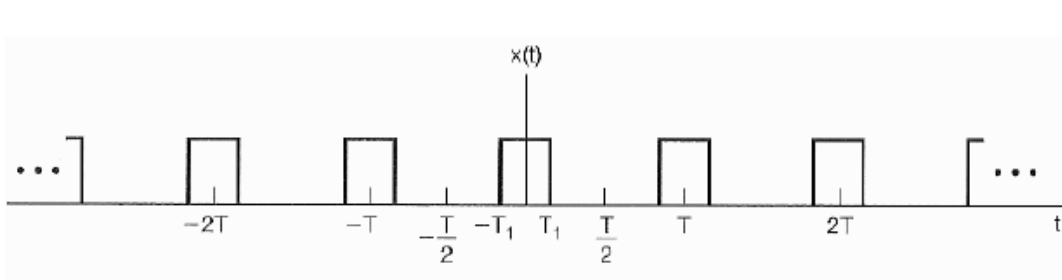
Continuous-time Fourier Transform



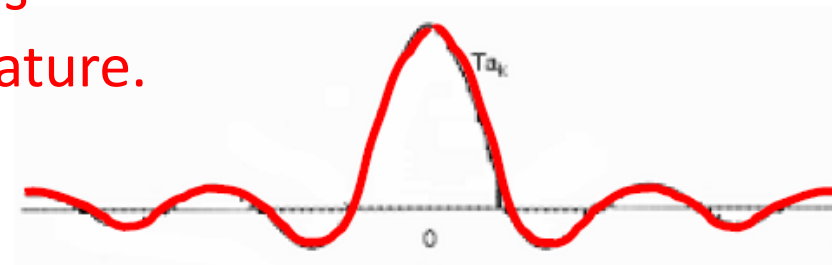
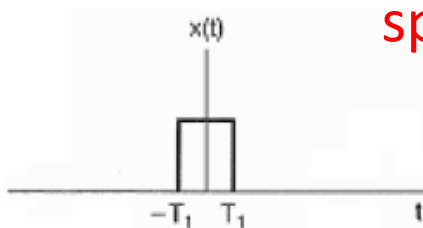
- If we increase the fundamental time period T of signal, the separation between spectral lines reduces.

Continuous-time Fourier Transform

$$T \rightarrow \infty \Rightarrow \{Ta_k\} \rightarrow \left. \frac{2 \sin(\omega T_1)}{\omega} \right|_{\omega = k\omega_0}$$



- Aperiodic signals has continuous spectrum in nature.



Continuous-time Fourier Transform

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_o t} \quad \text{Synthesis Equation of FS}$$

$$c_n = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jn\omega_o t} dt \quad \text{Analysis Coefficient of FS}$$

$$Tc_n = \int_{-T/2}^{T/2} x(t) e^{-jn\omega_o t} dt$$

$$X(\omega) = \lim_{T \rightarrow \infty} Tc_n = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad \text{As } T \rightarrow \infty, n\omega_o \rightarrow \omega$$

The above equation is referred to as **Fourier transform analysis equation** (or **forward Fourier transform equation**).

Continuous-time Fourier Transform

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t} \quad \text{Synthesis Equation of FS}$$

$$x(t) = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{n=-\infty}^{\infty} T c_n e^{jn\omega_0 t}$$

$$x(t) = \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} df$$

$$\text{As } T \rightarrow \infty, n\omega_0 \rightarrow \omega, \frac{1}{T} \rightarrow df$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

$$\text{As } d\omega = 2\pi df$$

The above equations is referred to as the **Fourier transform synthesis equation** (or **inverse Fourier transform equation**).